Autumn 2025 | **Physics 141: Exam IV Part II** | University of Rochester

Professor Sowinski | *Department of Physics & Astronomy*

Instructions

There is one question on this exam.
You have 4.8×10^4 min to catch up to 12 🧿's. That's 4×10^3 min per 🧿.
No partial derivatives are need on this exam, but you'll see them soon.
Draw arrows in the sand if you visit a beach like → 👆.
Accompany them with labeled and neat diagrams.
Dream legibly with a font size greater than 50.
🔱 May the Force be with you.🔱

On my honor and reputation as a scholar of Natural Science,
I will neither give nor receive unauthorized help on this exam,
and hereby vow that all work will be my own.

Signature/Print _____

Points

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🧿🧿🧿🧿
🧿🧿🧿🧿

$G = 6.674 \times 10^{-11} \frac{\mathrm{m}^3}{\mathrm{kg} \cdot \mathrm{s}^2}$ $c = 2.998 \times 10^8 \frac{\mathrm{m}}{\mathrm{s}}$ $h = 6.626 \times 10^{-34} \mathrm{J} \cdot \mathrm{s}$ $k_B = 1.381 \times 10^{-23} \frac{\mathrm{kg} \cdot \mathrm{m}^2}{\mathrm{s}^2 \cdot \mathrm{K}}$

$M_\oplus = 5.972 \times 10^{24}\mathrm{kg}$ $M_\odot = 1.989 \times 10^{30}\mathrm{kg}$ $R_\oplus = 6.371 \times 10^6\mathrm{m}$ $R_\odot = 6.957 \times 10^8\mathrm{m}$

$AU = 1.496 \times 10^{11}\mathrm{m}$ $M_☾ = 7.348 \times 10^{22}\mathrm{kg}$ $R_☾ = 1737\mathrm{km}$ $r_{☾-\oplus} = 3.844 \times 10^5\mathrm{km}$

	Cartesian Coordinates	Polar Coordinates	Cartesian $\Leftrightarrow$	Polar
	$\boldsymbol{r} = x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}}$	$\boldsymbol{r} = r\hat{\boldsymbol{r}}$	$x = r\cos\theta$	$r = \sqrt{x^2+y^2}$
$\boldsymbol{v} = \frac{d\boldsymbol{r}}{dt} = \dot{\boldsymbol{r}}$	$\dot{\boldsymbol{r}} = \dot{x}\hat{\boldsymbol{\imath}} + \dot{y}\hat{\boldsymbol{\jmath}}$	$\dot{\boldsymbol{r}} = \dot{r}\hat{\boldsymbol{r}} + r\dot{\theta}\hat{\boldsymbol{\theta}}$	$y = r\sin\theta$	$\theta = \arctan\frac{y}{x}$
$\boldsymbol{a} = \frac{d\boldsymbol{v}}{dt} = \dot{\boldsymbol{v}}$	$\ddot{\boldsymbol{r}} = \ddot{x}\hat{\boldsymbol{\imath}} + \ddot{y}\hat{\boldsymbol{\jmath}}$	$\ddot{\boldsymbol{r}} = (\ddot{r} - r\dot{\theta}^2)\hat{\boldsymbol{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\boldsymbol{\theta}}$		

	Inertial Frame	Rotating Frame
	$\boldsymbol{v} = \frac{d\boldsymbol{r}}{dt} = \dot{\boldsymbol{r}}$	$\boldsymbol{v} = \boldsymbol{v}_{\mathrm{rot}} + \boldsymbol{\Omega} \times \boldsymbol{r}$
	$\boldsymbol{a} = \frac{d\boldsymbol{v}}{dt} = \dot{\boldsymbol{v}}$	$\boldsymbol{a} = \boldsymbol{a}_{\mathrm{rot}} + 2\boldsymbol{\Omega} \times \boldsymbol{v}_{\mathrm{rot}} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \boldsymbol{r})$

Newton's 2nd Law :	$\boldsymbol{F} = \frac{d\boldsymbol{p}}{dt}$	Momentum :	$\boldsymbol{p} = m\boldsymbol{v}$
Impulse :	$\Delta\boldsymbol{p} = \int \boldsymbol{F}dt$	Universal Gravitation :	$\boldsymbol{F}_g = -G\frac{m_1 m_2}{\lvert\boldsymbol{r}_2 - \boldsymbol{r}_1\rvert^3}(\boldsymbol{r}_2 - \boldsymbol{r}_1)$
Work :	$W = \int_C \boldsymbol{F}(\boldsymbol{r}) \cdot d\boldsymbol{r}$	Kinetic Energy:	$K = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$
Work-Energy Theorem:	$W = \Delta K$	Conservative Force :	$\boldsymbol{F} = -\nabla U$
Spring Potential Energy :	$U_s = \frac{1}{2}m\omega^2\lvert\boldsymbol{r} - \boldsymbol{r}_{eq}\rvert^2$	Potential Energy :	$U(\boldsymbol{r}_f) - U(\boldsymbol{r}_i) = -\int_C \boldsymbol{F} \cdot d\boldsymbol{r}$
Grav. Potential Energy :	$U_g = -G\frac{m_1 m_2}{\lvert\boldsymbol{r}_2 - \boldsymbol{r}_1\rvert}$	Mass of object :	$M = \int_\Omega \rho(\boldsymbol{r})dV$
Center of mass:	$\boldsymbol{R} = \frac{1}{M}\sum_i m_i\boldsymbol{r}_i$ or	$\frac{1}{M}\int_\Omega \rho(\boldsymbol{r})\boldsymbol{r}dV$	
Angular Momentum:	$\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p} = I\boldsymbol{\omega}$	Torque:	$\boldsymbol{\tau} = \boldsymbol{r} \times \boldsymbol{F} = I\boldsymbol{\alpha}$
Moment of Inertia:	$I = \sum_i m_i(r_i^\perp)^2$		

$$\boldsymbol{A} \cdot \boldsymbol{B} = \lvert\boldsymbol{A}\rvert\lvert\boldsymbol{B}\rvert\cos\theta \quad \lvert\boldsymbol{A} \times \boldsymbol{B}\rvert = \lvert\boldsymbol{A}\rvert\lvert\boldsymbol{B}\rvert\sin\theta \quad \boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = \boldsymbol{B}(\boldsymbol{A} \cdot \boldsymbol{C}) - \boldsymbol{C}(\boldsymbol{A} \cdot \boldsymbol{B})$$

Expanding δx around x $\quad f(x + \delta x) = f(x) + f'(x)\delta x + \frac{1}{2!}f''(x)\delta x^2 + \cdots$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} - \cdots \qquad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \qquad \ln\lvert 1 + x\rvert = x - \frac{x^2}{2} + \frac{x^3}{3} + \cdots$$

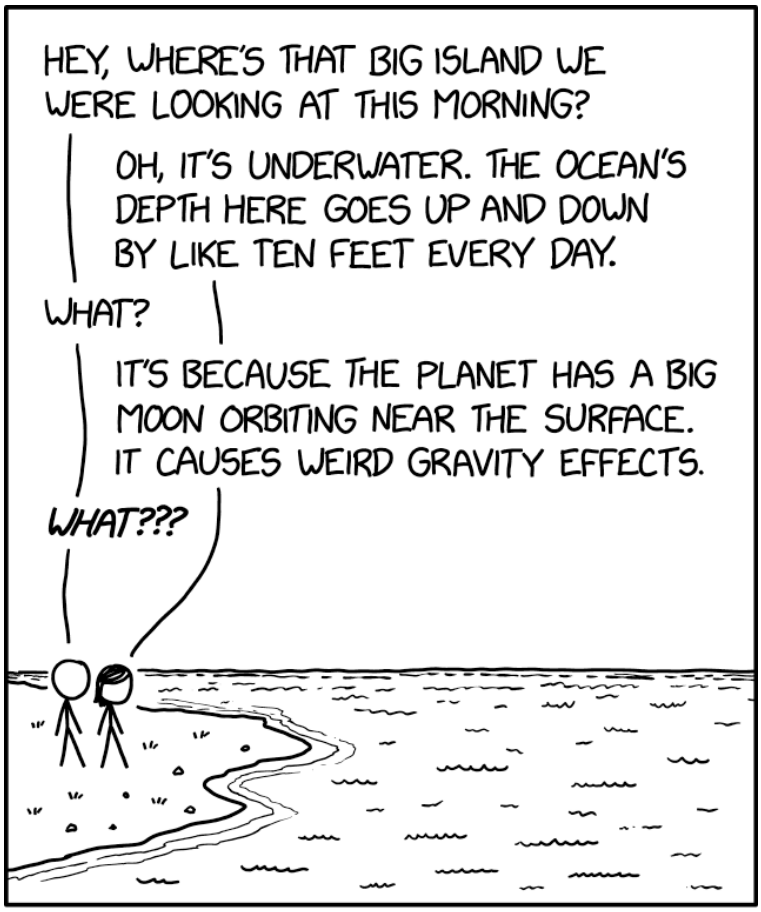
$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2}x^2 + \cdots \qquad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \qquad \frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$$

$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \sin\beta\cos\alpha$ $\quad \cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\beta\sin\alpha$

$\sin 2\theta = 2\sin\theta\cos\theta$ $\quad \cos 2\theta = \cos^2\theta - \sin^2\theta$ $\quad \cos\frac{\theta}{2} = \sqrt{\frac{1+\cos\theta}{2}}$ $\quad \sin\frac{\theta}{2} = \sqrt{\frac{1-\cos\theta}{2}}$

Physical Challenge

(🔥🔥🔥🔥🔥🔥🔥🔥🔥🔥🔥🔥🔥) You did it! You finished your first semester! Have an awesome Winter Solstice, and a relaxing Winter Break!



PEOPLE HERE ARE USED TO THEM, BUT TIDES ARE ONE OF THE WEIRDEST AND MOST SCI-FI ELEMENTS OF LIFE ON EARTH.